STAT4C03/6C03 Generalized Linear Models

Assignment 4 Due at 10pm on Thursday March 26, 2026

Instructions:

1. The only way to submit your assignment is through the Crowdmark page for the course. Use the link on Avenue (Content > Crowdmark) to login to Crowdmark and follow the instructions there to upload your assignment.

2. It is your responsibility to ensure that your submitted solutions are readable and complete. Any submitted file which cannot be opened will be given a mark of 0.

3. You may use R Markdown or Knitr to compile the R questions in your assignment but are not required to do so. For any question using R, you are required to include both the R code and output in your solutions as well as your written comments on the model(s) fitted.

4. Late assignments will be automatically assessed a penalty of 2% per hour, or part thereof, that they are late according to the Crowdmark time stamp.

5. You may update your submission up until the deadline. Once the due date has passed, you will no longer be able to update any submission. This includes adding questions that you did not submit before the deadline.

6. Please indicate **clearly** on your solutions whether you are in STATS 4C03 or STATS 6C03.

7. You are expected to show all details of your solution and any results taken from my notes or the textbooks must be clearly and properly referenced.

8. Students with accommodations through Student Accessibility Services who require accommodations for this assignment must contact the professor **at least 48 hours before the assignment is due** to request any such accommodations.

9. No other extensions to the due date and time will be given except in extreme circumstances.

10. There are 100 marks available for STATS 4C03 students and 125 for STATS 6C03 students. Note that the questions are not equally weighted. Marks for each question are given at the start of the question.

Q. 1 [30 marks]

The dataset in `Quine.csv` (slightly edited from `quine` in the `MASS` package) contains data on days absent from school in one year for students in New South Wales, Australia from a 1975 Australian National University PhD thesis by S. Quine. The covariates are whether the students were aboriginal or not, their sex, their grade (6–9), and whether they are classified as an average or slower than average learner.

a) Fit a model relating the number of days absent to the covariates including all two-way interactions. Give the summary output of the model and explain any unusual features. Does the model fit well?

b) Examine diagnostic plots for the fitted model. Do they show any issues with the model?

c) Apply `drop1` to the model and comment on the results.

d) For each combination of the categorical covariates, find the mean and variances of the absentee days and plot them against each other and comment on what the plot tells us.

e) An alternative approach could be to use a quasi-Poisson model to account for overdispersion. Fit such a model and conduct the `drop1` analysis again. Comment on your results and any differences from the earlier analysis.

f) For both the Poisson and quasi-Poisson models, fit a model which only has interactions that are significant in the `drop1` analysis and compare to the model fitted earlier.

Q. 2 [30 marks]

In this question we shall revisit the Berekley admissions data we examined in the third assignment. The data are in the file `Admission.csv`.

a) Consider the $2 \times 2 \times 6$ contingency table determined by the admission status, gender and department. Show that the best association structure for this dataset is that gender and admission status are conditionally independent given the department.

b) Since the primary response variable in the example is the admission status we could model this using a logistic regression with binomial response. Fit a main effect binomial logistic model. What do the regression parameters of this model represent? Write a report to summarize what the estimates and Wald tests for the individual parameters tell us about this fitted model.

c) Show that the binomial model fitted in part (b) is equivalent to a log-linear (Poisson) model with all 2-way interactions. That is, show that the two models result in exactly the same fitted counts.

d) Within each department find the ratio of the fitted odds of admission for the two genders (odds of admission for females divided by odds of admission for males) for the model fitted in (c). What do you notice about these 6 odds ratios?

e) Now consider the conditional independence model fitted in part (a) and calculate the same 6 fitted odds ratios within department for this model. How do they compare to the odds ratios in part (d)?

Q. 3 [40 marks]

The dataset `HSB.csv` (also available as `hsb` in the `faraway` package) contains data taken from the "High School and Beyond" study conducted by the National Education Longitudinal Studies (NELS) program of the National Center for Education Statistics (NCES) in the USA. The question of interest was to see what factors influence the choice of high school program (Academic, Vocational or General).

a) Make a table showing the proportion of males and females choosing the three different programs and comment on any differences. Repeat this comparison using socioeconomic status (column `SES`) rather than gender.

b) Divide the reading score into 4 ranges using the quartiles and make a plot of the observed distribution of program choice in each of the 4 ranges. Use lines to join the proportions for each program across the 4 ranges. Repeat this process for the math score and comment on both plots.

c) Fit a multinomial model to estimate the probability of each program choice using all other covariates in the model. Examine the estimated covariates for the 5 subject scores. Are any of them unexpected or inconsistent?

d) Interpret the results of your model for gender, math score and science score.

e) Construct a new variable which is the sum of the 5 subject scores and replace the individual subject scores in your model with this new. Interpret the estimated effects for this score on the subject choice.

f) Explain why the model you fitted in (e) is a nested model relative to the one fitted in (c) and conduct a likelihood ratio test to compare these two models. Which of these models would you prefer and why?

g) Use your chosen model to predict the program for each student in the dataset and compare these to the actual programs. In what proportion are the predicted and actual outcomes the same?

Q. 4 [Stats 6C03 only. 25 marks]

In Question 1, you used a quasi-Poisson approach to modelling overdispersion. An alternative is to use the negative binomial distribution. A convenient form for the probability mass function of this distribution is

$$\mathrm{P}(Y = y; \mu, \theta) \;=\; \frac{\Gamma(\theta + y)}{y!\Gamma(\theta)} \left(\frac{\theta}{\theta + \mu}\right)^{\theta} \left(\frac{\mu}{\theta + \mu}\right)^{y} \qquad y = 0, 1, 2, \ldots$$

where the parameters are $\mu > 0$ and $\theta > 0$. Note that when θ is a positive integer, it represents the number of successes required (usually labelled r) but this generalization allows for non-integer values of θ also.

a) Show that the mean and variance of this distribution are

$$\mathrm{E}(Y) \;=\; \mu \quad \text{and} \quad \mathrm{Var}(Y) \;=\; \mu + \frac{\mu^2}{\theta}$$

and so the variance is greater than the mean.
Hint: For any $\theta > 0$ and $0 < p < 1$

$$\sum_{y=0}^{\infty} \frac{\Gamma(\theta + y)}{y!\Gamma(\theta)} p^y \;=\; (1 - p)^{-\theta}$$

b) The `MASS` library provides a `negative.binomial` family which fits a negative binomial model with a fixed θ which is passed as an argument to the function. Repeat the plot done in 1(d) but now add the curves $v = m + m^2/\theta$ to the plot for $\theta = 1$, $\theta = 2$ and for $\theta = 3$. Which of these values of θ do you think best fits the data? Fit that model to the `Quine.csv` data and comment on the results of applying `drop1`.

c) Deciding on the correct θ is difficult so it may be preferable to think of θ as a parameter and estimate it using maximum likelihood. The function `glm.nb` in the `MASS` package does this. Use this function to fit the model to the `Quine.csv` data. Are there any differences for the `drop1` analysis? Conduct a Wald test of $H_0 : \theta = 2$ based on your fitted model from `glm.nb`.